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$$\lim_{x \rightarrow 0} \left(\frac{1}{\ln(1+x)} - \frac{1}{x} \right) = \frac{1}{0} - \frac{1}{0} = \infty - \infty$$

$$\lim_{x \rightarrow 0} \frac{x - \ln(1+x)}{x \ln(1+x)}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{(x - \ln(1+x))'}{(x \ln(1+x))'}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \frac{1}{1+x}}{\ln(1+x) + \frac{x}{1+x}}$$

$$= \lim_{x \rightarrow 0} \frac{x}{(1+x) \ln(1+x) + x} = \frac{0}{0}$$

Terug L'Hopital:

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{x'}{((1+x) \ln(1+x) + x)'}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\ln(1+x) + (1+x) \frac{1}{1+x} + 1}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\ln(1+x) + 2} = \frac{1}{\ln(1) + 2} = \frac{1}{2}$$

References